

Database and Local Storage Specification

Media Reminder - Offline-First Adaptive Media Alarm

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Contents

0.1 Document control

Field	Value
Document ID	MR-09
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Purpose	Define the Room schema, relationships, constraints, migrations, file layout, integrity model and transactional recovery.

Reading rule: This pack specifies a production-oriented Android application, not a promise that third-party devices will behave identically. Where Android or an OEM controls presentation, timing, sound, or permissions, the app provides transparent status and the strongest compliant fallback.

0.2 Document conventions

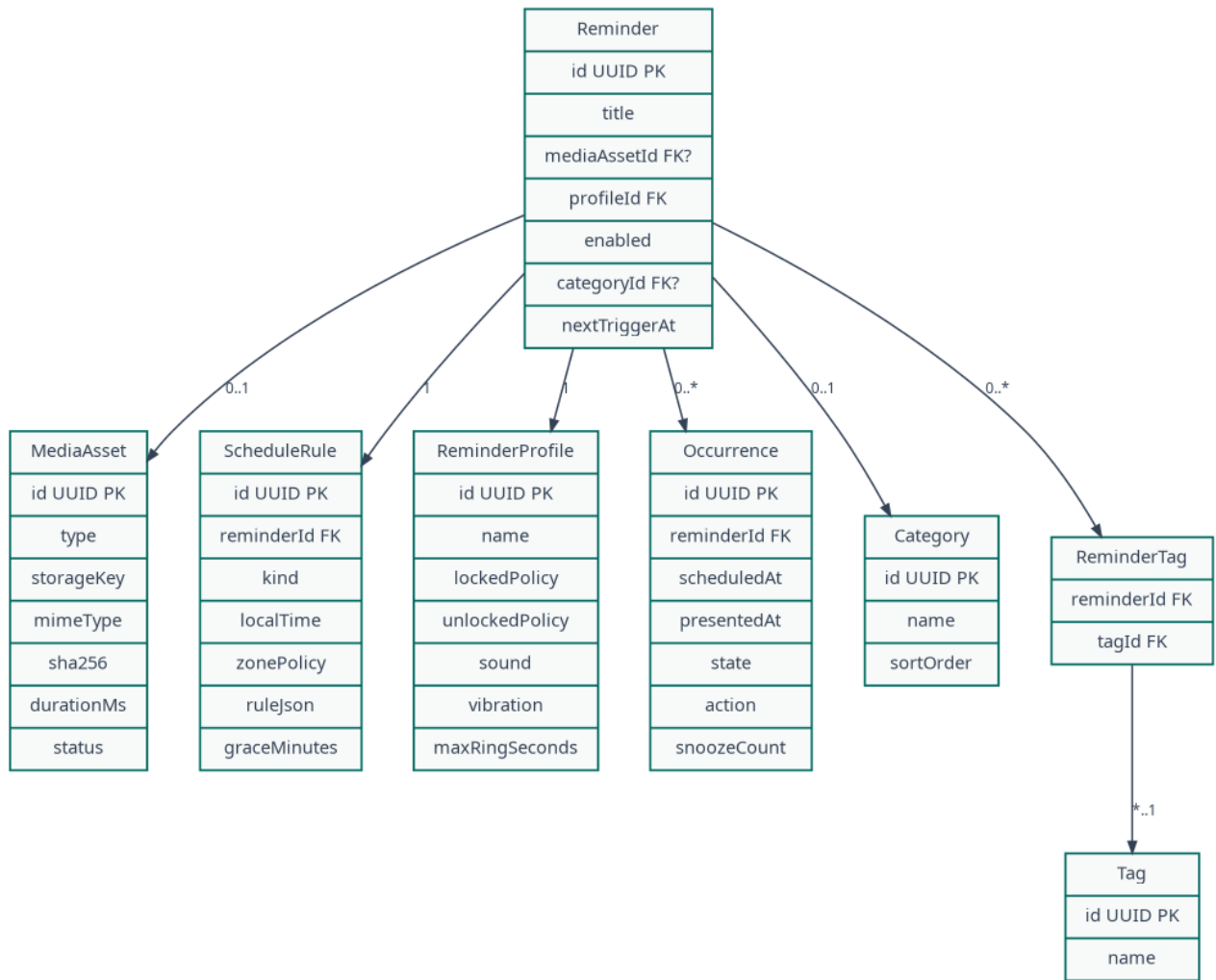
The words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are normative. A requirement ID is stable once published. Requirement IDs may be retired but **MUST NOT** be reassigned to a different meaning.

Term	Meaning
Reminder	A user-authored instruction that connects content, a schedule, and a presentation profile.
Occurrence	One calculated due instance of a reminder.
Alarm session	The bounded runtime state created when an occurrence is actively alerting.

Term	Meaning
Media asset	An app-owned local video, audio, image, or text item.
Profile	Reusable alert behavior such as Gentle, Standard, or Persistent.
Exact-alarm access	Android special app access that allows exact scheduling where the platform requires it.
Full-screen intent	Android notification mechanism for urgent, time-sensitive activity presentation. It is not a general overlay.
Heads-up notification	System-rendered high-priority notification shown temporarily over the current app when Android permits it.
Source of truth	The authoritative specification or local persisted record for a decision or state.

1 Persistence objectives

Persistence MUST remain correct through process death, reboot, partial file copy, notification action races, import cancellation and schema upgrades. Room is authoritative for logical data. The filesystem stores app-owned bytes. DataStore stores small preferences and a device-protected alarm envelope.



2 Database configuration

- Database filename is internal and never included in backups.
- Room uses WAL mode where supported.
- Foreign keys are enabled.
- Destructive migration is prohibited in release builds.
- Encryption-at-rest beyond Android sandbox/file-based encryption is a future option; v1 backups are explicitly unencrypted.
- Timestamps are UTC epoch milliseconds in Room; bridge/export serializes ISO-8601.
- Primary IDs are UUID strings generated by secure platform UUID.
- User-visible sort normalization is stored separately only when profiling demonstrates need.

3 Core entities

3.1 media_assets

Column	Type	Constraints/purpose
id	TEXT	PK UUID

Column	Type	Constraints/purpose
kind	TEXT	video/audio/image/text enum
title	TEXT	1-160 Unicode scalar length
notes	TEXT nullable	max 4000 characters
storage_key	TEXT	relative opaque filename, unique
mime_type	TEXT	normalized detected type
size_bytes	INTEGER	nonnegative
sha256	TEXT	64 lowercase hex, indexed
duration_ms	INTEGER nullable	nonnegative
width_px,height_px	INTEGER nullable	decoded metadata
category_id	TEXT nullable	FK categories, SET NULL
integrity_state	TEXT	healthy/unchecked/missing/changed/unsupported
created_at,updated_at	INTEGER	UTC millis
entity_version	INTEGER	optimistic concurrency

No source gallery URI is required after copy. Optional provenance such as original display name is excluded by default or stored as user-visible metadata only after privacy review.

3.2 reminder_profiles

Stable built-in UUIDs are seeded in migration. Columns include name, built-in flag, channel class/version, sound mode/reference, vibration JSON validated by converter, full-screen-on-locked Boolean, timeout seconds, retry count/interval, grace seconds, default snooze minutes, created/updated and entity version.

Profile timeout range: 15-600 seconds. Retry count: 0-3 in v1. Vibration total duration is capped.

3.3 reminders

Column	Purpose
id	PK UUID
media_id	FK media, restrict until dependency policy chosen
profile_id	FK profile, restrict
label	display label; default copied from title but independent
notes	reminder-specific notes
enabled_intent	user's desired enabled state
effective_state	disabled/needs_setup/active/archived
snooze_policy_json	validated bounded values

Column	Purpose
history_enabled	local recording preference
created_at,updated_at	timestamps
entity_version	optimistic concurrency

Effective state is derived but persisted for fast native queries and backup transparency; reconciliation can recompute it.

3.4 `schedule_rules`

One-to-one with reminder in v1. Fields include type, once instant, local time seconds-of-day, ISO weekday mask, zone policy, origin zone, DST gap policy, DST overlap policy and rule version. Check constraints ensure only relevant fields are set for each type.

3.5 `occurrences`

Column	Purpose
id	PK UUID
reminder_id	FK reminder cascade
kind	base/snooze/retry/test
parent_occurrence_id	nullable self FK
scheduled_at	due instant
occurrence_key	unique deterministic key for base occurrence
state	pending/claimed/accepted/snoozed/ dismissed/missed/timed_out/ failed_safe
triggered_at,resolved_at	nullable timestamps
action	terminal action code
created_at	timestamp

Unique (reminder_id, occurrence_key) prevents recurrence duplication. Only a bounded horizon of future derived occurrences is stored; the scheduler needs the next one, not years of rows.

3.6 `active_alarm_session`

At most one row is active, enforced by a singleton key or unique partial invariant in repository logic. Contains session UUID, occurrence UUID, state, started time, presentation decision, notification ID, service started flag, wake deadline, action nonce generation and last update. Terminal sessions are moved to diagnostics/history or deleted after reconciliation.

3.7 `scheduler_state`

Singleton row: desired occurrence UUID/time, desired generation, applied generation, pending-intent identity, last reconcile time, last reason, last error code. It is an outbox record for AlarmManager synchronization.

3.8 `pending_file_operations`

Tracks operation UUID, kind, temporary relative path, target relative path, associated entity UUID, phase, expected bytes/hash, created/updated and last error. Startup repair uses it to delete or finalize safe operations.

3.9 idempotency_records

Stores scope, key, request hash, result summary, created/expiry. Alarm action records persist long enough to cover repeated notification intents. General UI operation records have shorter retention.

3.10 categories and tags

categories(id,name,normalized_name,sort_order,created_at,updated_at) with unique normalized name.

tags(id,name,normalized_name,created_at,updated_at) with unique normalized name.

reminder_tags(reminder_id,tag_id) and optionally media_tags(media_id,tag_id) use composite primary keys.

3.11 schema_meta

Stores schema version, backup writer version, created app version, last migration version/time and integrity check state.

4 Indices

Required indices:

- media_assets(sha256);
- media_assets(category_id, updated_at);
- reminders(effective_state, updated_at);
- occurrences(state, scheduled_at);
- occurrences(reminder_id, occurrence_key) unique;
- pending_file_operations(phase, updated_at);
- idempotency_records(scope, key) unique and expires_at;
- category/tag normalized names unique.

Query plans for Today, next occurrence, active reminder count and media dependency count are captured in performance tests.

5 File storage

Imported bytes use an opaque UUID filename and normalized extension. The mapping is internal:

files/media/7f3...a9.mp4

No user-supplied path segment enters the file path. Text assets are JSON or UTF-8 in a separate directory. Derived thumbnails are WebP cache and may be cleared at any time. Export archives are written to a user-selected document URI where possible; a temporary private file is only used when the destination API requires it.

6 Storage limits

Defaults, configurable only through build constants:

- individual asset soft warning at 500 MB;
- individual asset hard limit at 2 GB for v1 unless large-file tests pass;
- backup expected uncompressed hard limit at 10 GB or 80% of available space, whichever is lower;
- maintain 250 MB or 5% free-storage reserve after import/restore, whichever is greater;

- text title 160 characters, notes 4000, category/tag 60;
- diagnostics 5 MB ring buffer;
- thumbnails cache 250 MB with LRU eviction.

These values are product protections, not filesystem capabilities. The UI displays estimates before copy.

7 Import atomicity

The database and filesystem cannot share a true atomic transaction. Use a journaled two-phase protocol:

1. persist pending operation;
2. stream and verify temporary file;
3. atomically rename on same filesystem;
4. transaction inserts record and marks operation committed;
5. cleanup removes journal.

On startup:

- temp exists, no target, phase Copying -> delete temp;
- target exists, no media row, phase Promoted -> verify hash then complete row only if full metadata exists, otherwise quarantine/delete;
- media row exists, target missing -> mark Missing and disable reminders;
- both exist and hash valid -> complete/clear operation.

8 Backup transaction support

Before Replace import, current logical records are exported to a private rollback package and current media map is journaled. Commit stages new media under an operation directory, inserts data in a Room transaction using conflict plan, atomically promotes files, applies scheduler desired state and then removes rollback after verification. A crash at any point is recovered by operation phase.

9 Database migrations

Every schema version has:

- forward Room migration;
- migration fixture from every supported prior public version or a tested chain;
- invariant checks;
- backup reader compatibility note;
- rollback strategy at release level (usually reinstall + restore, not database downgrade).

Migrations MUST be idempotent where retry could occur and MUST NOT hash entire large files on the main thread. Built-in profile changes use stable IDs and explicit version migration, preserving user edits unless reset.

10 Integrity checks

Startup performs lightweight checks only. Full `PRAGMA integrity_check`, media hash audit and orphan scan are user-invoked or deferrable maintenance. Health reports last full check. Any database corruption path first copies the damaged database to a private recovery area with bounded retention, then offers Restore backup or Reset. It never silently deletes data.

11 Data retention

Occurrence history defaults to 90 days. Diagnostic event retention is size/time bounded. Idempotency records are retained by scope: alarm action 7 days, UI mutation 24 hours, backup commit 30 days. Completed export temporary files expire after 24 hours unless the user saved them externally.

12 Uninstall and clearing data

The app explains that Android removes app-specific media and database on uninstall or Clear storage. External exported ZIPs are not deleted. Cache clearing does not remove source media or reminders. "Reset app" requires explicit typed/destructive confirmation and optional export prompt.

13 Database acceptance

- Foreign-key and check constraints pass on all fixtures.
 - Crash injection at each two-phase file step reaches a documented repaired state.
 - No database column contains an absolute filesystem path.
 - Next occurrence query remains within MR-15 budget at 10,000 reminders/50,000 retained occurrences.
 - Migration tests cover every public schema version.
 - Export/import uses DTO mapping, not database file copy.
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13.1 Governance

This document is part of the **Media Reminder Source-of-Truth Pack v1.0**. In case of conflict, apply this precedence order: (1) explicit platform safety and permission rules, (2) Architecture Decision Records, (3) Android specification, (4) data and backup specifications, (5) PRD and feature specification, (6) UX and visual guidance, (7) implementation notes. Record any intentional deviation in the decision log before release.